



Extensible Refinement Layer for LLM-Based Checker Generation

NORM: merge · de-dup · keep · missing

KNighter or other
LLM generator

(a) LLM Generator

context

Draft
checker/query



Patch



Source tree



Validation target



(b) Evidence assembly (typed)



Patch preflight

patch_fact, requirements, mechanism graph



Source extraction

semantic_slice: function, guards,
calls, mem/state, build flags



Analyzer-native

CSA: path_guard, state_transition, lifetime
CodeQL: dataflow_candidate, call_chain



Validation feedback

validation_outcome:
failure_mode, repair

(c) EvidenceBundle

records = ⟨type, analyzer,
scope, location, payload,
provenance, slice⟩EvidenceSlice:
statements, guards, call_edges,
state_transitions,
api_terms, related_symbols,
coverage_status
missing_evidence
kept · collected_analyzers

(e) Audit artifacts

evidence bundle
validation feedback
refinement input
analyzer results
final report

(d) Refinement plugin (target-preserving)



Refinement
agent

request types



Candidate
checker/query

persist



Strict gate

accept if Changed
 $\wedge \text{AnalyzerOK} \wedge \text{PDS (bug hit} \wedge$
fix silent)}

adopt

feedback